AIR POLLUTION AND CONTROL

Subject Code : 23A01504b

UNIT - V : AUTOMOBILE AND INDOOR POLLUTION

Syllabus: Vehicular pollution - Sources and types of emission - Effect of operating conditions - Alternate fuels and emissions - Emission controls and standards, Strategies to control automobile pollution - Causes of indoor air pollution - Changes in indoor air quality - Control and air cleaning systems - Indoor air quality.

PART A : AUTOMOBILE (VEHICULAR) POLLUTION

I. Vehicular Pollution - Introduction

Road vehicles are **mobile, ground-level, line sources** that release pollutants close to people (breathing height), so their effect is much greater than the quantity suggests. In Indian metros vehicles contribute **30 - 70%** of CO, NO_x, HC and a major part of PM. Growth in vehicle numbers (> 30 crore registered), old vehicles, poor fuel quality, congestion and poor maintenance aggravate the problem.

- Fuel: petrol (spark ignition, SI engine, Otto cycle) and diesel (compression ignition, CI engine); two-stroke and four-stroke engines
- Complete combustion: $C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$
- Real engines give incomplete combustion: CO, HC, soot, NO_x, aldehydes in addition

2. Sources and Types of Emission

2.1 Sources of emission from a petrol vehicle

Source	Share of HC	Pollutants
Exhaust pipe (tailpipe)	about 60%	CO, HC, NO_x, aldehydes, Pb, particulates, SO_2
Crankcase blow-by	about 20%	Unburnt HC leaking past piston rings
Evaporative (fuel tank, carburettor)	about 20%	HC vapours (diurnal + hot soak + running losses)
Tyre, brake wear, road dust	-	PM (non-exhaust), heavy metals

2.2 Pollutants and how they form

Pollutant	Formation	Effects
Carbon monoxide (CO)	Incomplete combustion, rich mixture, low O_2; idling	Binds Hb; headache, death
Hydrocarbons (HC)	Unburnt fuel; flame quenching at cold walls; crevice volume; misfire	Smog, cancer (benzene, PAH)
NO_x (NO, NO_2)	Thermal (Zeldovich) at > 1800 K, excess O_2: $N_2 + O \rightarrow NO + N$; $N + O_2 \rightarrow NO + O$	Acid rain, smog, lung damage
Particulates (soot)	Diesel: fuel-rich pockets; cracking of fuel; also lube oil	Lung cancer; black smoke
Lead (Pb)	Tetra-ethyl lead anti-knock in petrol (phased out in India 2000)	Neurotoxic
SO_2	Sulphur in fuel (diesel > petrol)	Respiratory, acid rain
CO_2	Complete combustion; ~ 2.3 kg per litre petrol	Global warming
Aldehydes, PAH, benzene	Partial combustion, alcohol fuels	Irritant, carcinogenic

2.3 Petrol versus diesel engine emissions

Pollutant	Petrol (SI)	Diesel (CI)
CO	High (1 - 10%)	Low (< 0.5%) - lean mixture
HC	Medium - high	Low (but smell, aldehydes)
NO_x	Medium	High (high compression ratio, excess air)
PM / smoke	Very low	**High** - black smoke
SO_2	Low	Higher
CO_2	Higher per km	Lower (better efficiency)

Two-stroke engines (two wheelers, autos) emit 3 - 5 times more HC and PM than four-stroke because fresh charge escapes with exhaust (scavenging loss) and lubricating oil is burnt with the fuel (white smoke).

3. Effect of Operating Conditions

Emissions change with engine load, speed and especially **air-fuel (A/F) ratio**.

Stoichiometric A/F for petrol is about **14.7 : 1** by mass.

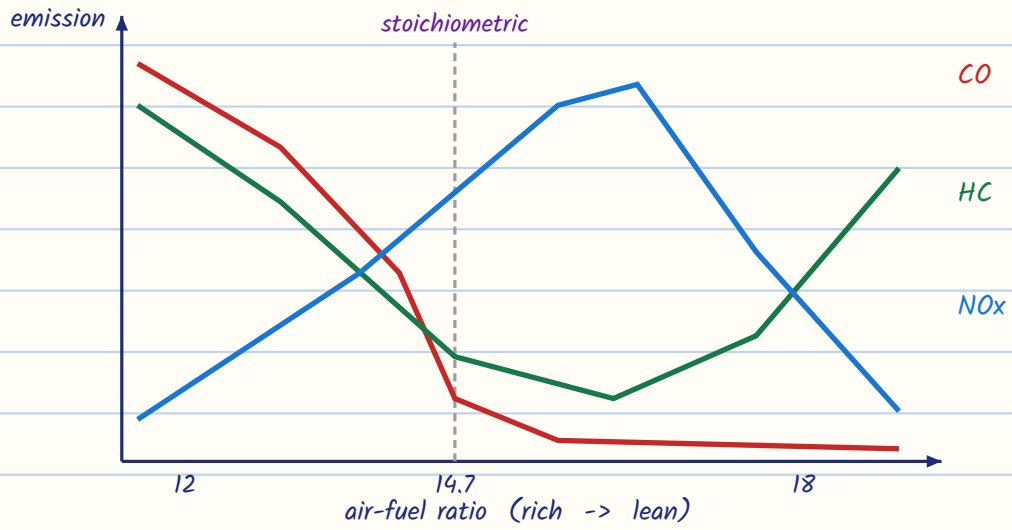


Fig. Variation of CO, HC and NO_x emissions with air-fuel ratio (SI engine)

Mode	Mixture / condition	CO	HC	NO_x
Idling	Rich, low temperature, poor mixing	Very high	High	Low
Acceleration	Rich, then high temperature and load	High	Medium	High

Mode	Mixture / condition	CO	HC	NO_x
Cruising (steady speed)	Lean, stable combustion	Low	Low	Medium - High
Deceleration	Throttle closed, high manifold vacuum; fuel film evaporates, misfire	Medium	Very high	Low
Cold start	Choke on; rich; catalyst not yet hot	Very high	Very high	Low

- **Rich mixture (A/F < 14.7):** lack of O_2; CO and HC increase sharply, NO_x is low (lower temperature)
- **Slightly lean (about 16 - 17):** maximum flame temperature -> **NO_x peak**; CO and HC are minimum
- **Very lean (> 18):** misfire, unstable combustion -> HC rise; NO_x falls (low temperature)
- **Ignition timing:** more advance -> more NO_x, less HC; retard reduces NO_x but increases fuel consumption
- **Compression ratio, engine temperature, load and speed:** all increase NO_x; high speed reduces HC and CO per km but raises NO_x
- **Engine maintenance, age, tuning, altitude, fuel quality** - worn rings, dirty air filter, leaking injectors increase CO and HC
- **Traffic congestion** (stop-go) lengthens idling and acceleration modes; emission per km at 20 km/h is 2 - 3 times that at 60 km/h
- **Diesel engines:** smoke increases with load, as A/F decreases at full load; NO_x increases with temperature and injection advance

4. Alternate Fuels and Emissions

Used to reduce emissions, dependence on crude oil and carbon footprint.

Fuel	Features	Emission effects
CNG (compressed natural gas, mainly CH_4, stored at 200 bar)	High octane (130), mixes well; cheaper; used in Delhi buses, autos	CO -90%, HC (non-methane) -, NO_x -, **no PM, no Pb, no SO_2**; CO_2 lower by 25%; methane is a GHG
LPG / Autogas (propane, butane)	Easy to liquefy at 8 bar; octane ~ 105	Lower CO, HC, NO_x than petrol; no PM
Ethanol (blends E10, E20 in India; bio-ethanol from sugarcane, grain)	Renewable, high octane; lower energy content; water-absorbing	Lower CO, HC; higher aldehyde (acetaldehyde); lower CO_2 life-cycle
Methanol (M85, M100)	From natural gas or coal / biomass; toxic; corrosive	Low NO_x, PM; formaldehyde emissions high
Biodiesel (FAME from jatropha, pongamia, used oil; BS - B20)	Renewable, biodegradable, sulphur-free, high cetane	Lower PM, CO, HC, SO_2; slightly **higher NO_x**
Hydrogen (IC engine / fuel cell)	Highest energy per kg; storage and cost problems	Only water in fuel cell; NO_x in IC engine; zero CO, HC, CO_2
Electric vehicles (EV) battery / hybrid / plug-in	Zero tailpipe emission; power from grid; FAME scheme	No local pollution; emissions shift to power plants unless renewable energy; battery disposal
Dimethyl ether (DME), producer gas, biogas, synthetic fuels	Diesel substitutes / blends	Low soot; limited use

CNG and EV are best for cities. BS-VI petrol and diesel (sulphur 10 ppm) enable after-treatment devices to work.

5. Emission Controls

5.1 Engine modifications (in-engine controls)

- **Better combustion chamber design** (hemispherical, swirl, squish), higher turbulence; **lean burn** engine; **stratified charge**
- **Electronic fuel injection** (MPFI, GDI, common-rail diesel) with ECU and **lambda (oxygen) sensor** closed-loop control instead of carburettor
- **Exhaust Gas Recirculation (EGR):** 5 - 15% exhaust gas is returned to the intake; inert gas lowers peak flame temperature -> **NO_x reduced by 50 - 70%**
- **Positive Crankcase Ventilation (PCV) valve:** blow-by gases returned to intake manifold -> crankcase HC removed
- **Evaporative emission control:** vapour from tank and carburettor is stored in an **activated carbon canister**; purged to intake when the engine runs
- **Turbocharging, intercooling, variable valve timing**, injection timing retard; high-pressure injection reduces smoke in diesel
- **Two-stroke:** direct injection (fuel injected after port closure) or conversion to four-stroke

5.2 Fuel modifications

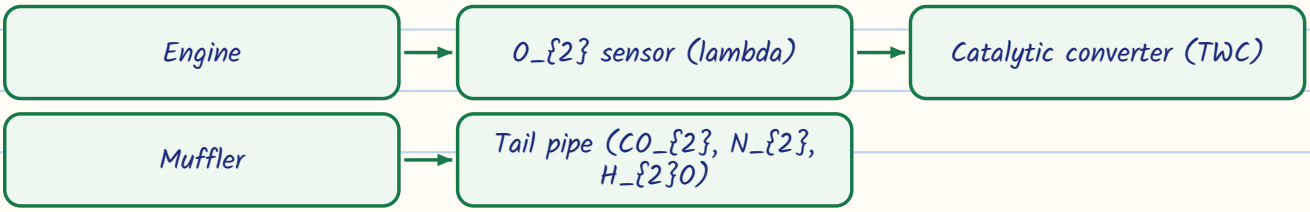
- Unleaded petrol (lead removed - allows catalyst); low-sulphur fuels (BS-VI: 10 ppm); low benzene (< 1%) and low aromatics; oxygenates (ethanol, MTBE) in blends; reformulated fuel; higher cetane diesel; fuel additives; clean alternative fuels

5.3 Exhaust after-treatment devices

Three-Way Catalytic Converter (TWC): a stainless steel shell with a ceramic honeycomb monolith (or metallic), coated with **alumina washcoat** impregnated with noble metals - **Pt, Pd (oxidation) and Rh (reduction)**. Operates near **stoichiometric** A/F (lambda = 1), temperature 300 - 800 deg C. Converts three pollutants:

- Oxidation of CO: $2\ CO + O_2 \rightarrow 2\ CO_2$
- Oxidation of HC: $C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$

- Reduction of NO_x: $2\ NO + 2\ CO \rightarrow N_2 + 2\ CO_2$; $2\ NO + 2\ H_2 \rightarrow N_2 + 2\ H_2O$
- Efficiency > 90% when hot; **poisoned by lead, sulphur, phosphorus** (so unleaded fuel needed); needs O_2 sensor



- **Diesel Oxidation Catalyst (DOC):** oxidises CO, HC and soluble organic fraction of PM
- **Diesel Particulate Filter (DPF):** wall-flow ceramic filter traps > 95% of soot; regenerated by burning soot at 550 - 600 deg C (active / passive)
- **SCR with urea (AdBlue / DEF):** 32.5% urea solution injected into the exhaust; NH_3 reduces NO_x to N_2 on catalyst (heavy diesel vehicles; BS-VI)
- **Lean NO_x trap, thermal reactor / secondary air injection** (older engines), **OBD (On-Board Diagnostics)** monitors the control system
- **Inspection and maintenance (I/M):** PUC (Pollution Under Control) certificate - CO and HC test (petrol), smoke opacity test (diesel, Hartridge units)

6. Emission Standards

Controlled in India under the **Motor Vehicles Act 1988, Central Motor Vehicles Rules 1989**, Air Act 1981 and Environment (Protection) Act 1986. **Bharat Stage (BS) norms** (introduced 2000) are based on European (Euro) norms.

Norm	Equivalent	Year (nationwide)
India 2000 / BS-I	Euro 1	2000 (all India)
BS-II	Euro 2	2001 (NCR), 2005 nationwide
BS-III	Euro 3	2005 (cities), 2010 nationwide
BS-IV	Euro 4	2010 (13 cities), 1 April 2017 nationwide
BS-V	Skipped	Leap-frogged to BS-VI
BS-VI	Euro 6	**1 April 2020** nationwide (phase-2 RDE from April 2023)

Passenger car limit (g/km)	BS-IV	BS-VI
Petrol: CO	1.0	1.0
Petrol: NO_x	0.08	0.06
Diesel: NO_x	0.25	0.08
Diesel: PM	0.025	0.0045
Fuel sulphur	50 ppm	10 ppm

- BS-VI introduced **DPF, SCR, OBD, GDI particle limits** (PM and particle number), real-driving emission test and evaporative control
- **Other:** noise limits; CAFE (fuel economy) norms; age limit and **vehicle scrappage policy**; mass emission standards for 2-3 wheelers; tests on chassis dynamometer using the Indian Driving Cycle / WLTP, with CVS sampling

7. Strategies to Control Automobile Pollution

Strategy	Measures
Technological	Catalytic converters, EFI, EGR, DPF; BS-VI norms; hybrid and electric vehicles; retrofitting of old buses; good maintenance
Fuel	Unleaded, low-sulphur, CNG / LPG, ethanol blend, biodiesel, hydrogen; fuel adulteration check
Traffic management	Synchronised signals, one-way roads, flyovers, bypass / ring roads, no-parking zones, speed limits, congestion pricing, odd-even scheme
Public transport	Metro rail, BRT, e-buses, last-mile connectivity, cycling and pedestrian lanes, car pooling
Urban planning	Mixed land use, green belt, plantation along roads (pollution sink), buffer zones, tall buildings away from roads
Legal / economic	PUC checks, age limit for vehicles, scrappage incentive, higher tax on old and diesel vehicles, EV subsidy (FAME), parking fee
Awareness	Drive less, switch off engine at red light, tyre pressure, correct driving

PART B : INDOOR AIR POLLUTION

8. Introduction to Indoor Air Quality

People spend **80 - 90% of their time indoors** (home, office, school, vehicles). **Indoor air pollution** is the presence of contaminants in the air inside buildings that may harm the health or comfort of occupants. Indoor levels of many pollutants are **2 - 5 times (sometimes 100 times) higher** than outdoors (US-EPA). The WHO estimates that **household air pollution** (cooking with solid fuel) causes about 3 million premature deaths per year, mostly in developing countries (women and children).

- **Indoor Air Quality (IAQ)** = the nature of the air that affects health and well-being of occupants
- **Sick Building Syndrome (SBS):** > 20% of occupants complain of headache, eye / nose / throat irritation, fatigue, dizziness, nausea - symptoms disappear on leaving the building, cause not specific
- **Building Related Illness (BRI):** a specific diagnosable disease from the building: Legionnaires' disease, hypersensitivity pneumonitis, asthma, humidifier fever

9. Causes of Indoor Air Pollution

Source	Pollutants	Health effects
Combustion - chulha, wood, dung, coal, kerosene, gas stove, heaters	CO, PM2.5, NO_2, SO_2, PAH, formaldehyde	COPD, lung cancer, ALRI in children, eye disease
Tobacco smoke (ETS)	PM, CO, nicotine, benzene, > 60 carcinogens	Cancer, asthma, heart disease
Building materials: asbestos, pressed wood, insulation, paint, adhesives	Asbestos fibres, formaldehyde, VOCs, Pb	Asbestosis, mesothelioma, irritation, cancer
Furnishing: carpets, foam, furniture, cleaning products, air-fresheners	VOCs (toluene, xylene), phthalates, ozone	Headache, allergic reaction
Soil / building stone	**Radon** (Rn-222, from decay of uranium - radium)	Lung cancer (2nd cause after smoking)
Biological: damp walls, AC coils, humidifiers, pets, carpets	Mould spores, dust mites, bacteria, Legionella, pollen, dander	Allergy, asthma, infections
Pesticides, mosquito coils, incense, candles	PM, VOC, pesticides	Neurological, respiratory
Office equipment: printers, photocopiers	Ozone, VOC, toner particles	Irritation
Occupants (breathing, perspiration)	CO_2, odours, bacteria, viruses	Stuffiness, infection spread
Infiltration of outdoor air	Traffic fumes, dust, pollen, industrial gases	Many

- **Poor ventilation** is the key factor: pollutants accumulate. Also high temperature and humidity, poor maintenance of HVAC and moisture problems (leaks, condensation)

10. Changes in Indoor Air Quality

- **Energy-efficient (tight) buildings** since 1970s oil crisis: sealed windows, low air-exchange rate (0.2 - 0.5 ACH) -> pollutants build up
- **Increase in synthetic materials** (plastics, composites, solvents, cleaners, electronics) -> more VOCs
- **Time of day / season:** cooking peaks, winter heating and closed windows give higher concentration; humidity in monsoon promotes mould
- **Urbanisation and outdoor pollution:** infiltration of PM2.5; **I/O ratio** (indoor / outdoor) shows if the source is indoor (> 1) or outdoor (< 1)
- **Changes in lifestyle:** more time indoors, AC (recirculation), air fresheners, new furniture ('new building smell')
- **Occupancy and activity:** CO_2 rises with number of people; CO_2 > 1000 ppm indicates inadequate ventilation (outdoor 400 ppm)
- **Chemical reactions indoors:** ozone + terpenes (cleaning fluids) -> secondary organic aerosol, formaldehyde

Mass balance model for a room (well-mixed)

$V\,dC/dt = Q\,C_0 - Q\,C + S - k\,V\,C$ -> steady state: $C = (Q\,C_0 + S) / (Q + k\,V)$

- V = room volume (m^3), Q = ventilation flow (m^3/h) = ACH x V, C_0 = outdoor conc., S = source emission rate (mg/h), k = removal / decay rate (1/h)
- **Example:** V = 50 m^3, ACH = 0.5 -> Q = 25 m^3/h; source S = 5 mg/h; C_0 = 0, k = 0 -> **$C = S / Q = 0.2$ mg/m^3**. Doubling ACH halves the concentration.

Pollutant	Guideline / limit (typical)
CO_2	1000 ppm (comfort indicator)
CO	9 ppm (8 hr); WHO 7 mg/m^3 (24 hr)
Formaldehyde	0.1 mg/m^3 (30 min, WHO)
Radon	100 Bq/m^3 reference level (WHO); action at 300
PM2.5 / PM10	WHO 24 hr: 15 / 45 ug/m^3
Temperature, RH	23 - 26 deg C, 30 - 60% (comfort)

Ventilation standards: ASHRAE 62.1 - about **7.5 L/s per person** (15 cfm) in offices; **National Building Code of India (NBC)**, IS 3362 (natural ventilation); IGBC / Green building guidelines for IAQ.

II. Control of Indoor Air Pollution and Air Cleaning Systems

II.1 Control strategies (in order of preference)

1. **Source control** (best): eliminate or reduce the source - no smoking indoors, **clean cooking fuel (LPG, biogas, electric, improved chulha with chimney)** - PM Ujjwala Yojana, low-VOC paints, formaldehyde-free boards, asbestos removal, store solvents outside, proper pest control
2. **Ventilation:** dilute the pollutants by outdoor air - natural (windows, vents, cross-ventilation), mechanical (exhaust fans, kitchen chimney / hood, bath exhaust), **heat recovery ventilator (HRV / ERV)**; ACH 0.5 - 1.0 for homes; offices 4 - 6 ACH; provide a 'flush-out' before occupying a new building
3. **Air cleaning:** remove pollutants from the air by equipment (when source control and ventilation are not enough)
4. **Control of humidity** (30 - 60%), fix leaks, cleaning, HVAC maintenance, filter change, **radon mitigation** (sealing cracks, sub-slab depressurisation fan), education of occupants

11.2 Air cleaning systems

Device	Principle	Removes	Remarks
Mechanical filters (flat, pleated, bag; MERV rating)	Impaction, interception, diffusion	Dust, pollen, mould	Cheap; clog; dP
HEPA filter	Fibre mat; 99.97% for 0.3 um	PM2.5, allergens, bacteria	High dP, costly; most used in homes, hospitals, clean rooms
Electronic air cleaner (ESP type)	Charge + collecting plates	Fine PM, smoke	Low dP; may emit ozone; wash plates
Activated carbon filter	Adsorption	VOC, odours, formaldehyde, ozone	Saturates; replace / regenerate
UVGI (UV-C light)	Kills micro-organisms by DNA damage	Bacteria, viruses, mould	Coil and duct disinfection; needs exposure time
Photocatalytic oxidation (TiO_2 + UV)	Oxidises VOC to CO_2 + H_2O	VOC, odour	By-products possible
Ionisers / ozone generators	Ions / ozone	Particles (ionisers)	**Not recommended** - O_3 is harmful
Botanical (plants), biofilters	Phytoremediation	Some VOC	Very small effect
Chemical / gas-phase ($KMnO_4$ on alumina)	Chemisorption	Acid gas, formaldehyde	Industrial

- **Clean air delivery rate (CADR)** = flow rate x efficiency (m^3/h) indicates cleaner capacity; choose CADR ~ 2/3 of the room area
- Portable cleaners work on re-circulated air: the effective removal rate k_c = CADR / V adds to the model above (C reduces by factor (Q + kV))

- Filters need regular maintenance; poor maintenance makes the filter a pollution source

12. Indoor Air Quality (IAQ) Management

- **Assessment:** walk-through inspection, occupant questionnaires, measurement of temperature, RH, CO_2, CO, PM2.5, TVOC, HCHO, radon, microbes (sampling with impactor, passive samplers, detectors)
- **IAQ standards:** WHO guidelines for indoor air quality; ASHRAE 62.1 & 55; Indian Green Building Council (IGBC) and the Bureau of Indian Standards (BIS) norms; Smoke-free policy
- **Management plan:** design (material selection, HVAC sizing, ventilation), commissioning, operation and maintenance, monitoring and complaint handling, communication
- Benefits: improved health, productivity and comfort; lower absenteeism, energy-saving through demand-controlled ventilation (CO_2 sensors)

13. Quick Revision / Important Questions

1. Explain the sources and types of emission from a petrol vehicle. Compare petrol and diesel pollutants.
2. Discuss the effect of operating conditions (A/F ratio, speed, load) on emission.
3. Write about alternate fuels (CNG, ethanol, biodiesel, hydrogen, EV) and their emissions.
4. Explain the three-way catalytic converter with a sketch. What are EGR and PCV?
5. What are Bharat Stage norms? List the strategies to control automobile pollution.
6. What are the causes of indoor air pollution? Explain SBS and BRI.
7. Explain the control methods and air-cleaning systems of indoor air quality (HEPA, carbon, UVGI).